

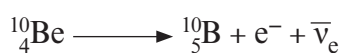
Answer all questions.

1. (a) Mesons, leptons and quarks are three groups of particles. State which of these groups are affected by:

(i) the weak nuclear force; [1]

(ii) the strong nuclear force. [1]

- (b) The following interaction shows a β^- decay for a beryllium nucleus.



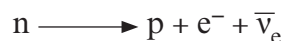
(i) State the name of the $\bar{\nu}_e$ particle. [1]

(ii) Explain how lepton number and charge are conserved in this interaction. [2]

- (c) (i) Write down the number of neutrons in:

I. ${}^{10}_4\text{Be}$ II. ${}^{10}_5\text{B}$ [1]

- (ii) On the level of baryons the interaction can be written as:



Show clearly that there is a change in quark flavour in this interaction. [2]

